

Part 1: Market Mechanics & Order Books

1. Types of Orders

- **Market Order:** Executes immediately at best available price (Taker).
- **Limit Order:** sits in the book at a specific price (Maker).

2. The Bid-Ask Spread

Key Formula

$$\text{Spread} = \text{Best Ask Price} - \text{Best Bid Price}$$

Exercise: Order Book Dynamics

Consider the following Limit Order Book for **NVIDIA (NVDA)**:

Bid (Buy Orders)		Ask (Sell Orders)	
Volume	Price	Price	Volume
100	\$100.00	\$100.05	500
200	\$99.95	\$100.10	200
50	\$99.90	\$100.15	100

- (a) What is the current Bid-Ask Spread?

Solution:

$$\text{Spread} = 100.05(\text{Best Ask}) - 100.00(\text{Best Bid}) = \$0.05$$

- (b) A trader submits a **Market Buy order for 600 shares**.

- How many shares are bought at what price?
- What is the weighted average price paid per share?

Solution: The Market Buy order consumes liquidity starting from the lowest Ask:

- **Step 1:** Buy all 500 shares at \$100.05. (Order remaining: 100).
- **Step 2:** Buy remaining 100 shares at \$100.10.

Calculation:

$$\text{Total Cost} = (500 \times 100.05) + (100 \times 100.10) = 50,025 + 10,010 = \$60,035$$

$$\text{Avg Price} = \frac{60,035}{600} \approx \$100.0583$$

- (c) After the trade in (b), what is the **new** Best Ask Price?

Solution: The level at \$100.05 is completely depleted (0 volume). The level at \$100.10 had 200 shares, we took 100, leaving 100 shares.

New Best Ask: \$100.10 (Volume: 100).

Part 2: Forward Contracts

Definitions

A **Forward Contract** is an agreement to buy (Long) or sell (Short) an asset at time T for price K .

Payoff Formulas at Maturity (T)

$$\text{Payoff for Long} = S_T - K$$

$$\text{Payoff for Short} = K - S_T$$

Example Problem (Preparation for Homework 1, Problem 1)

You enter into a **Short Forward Contract** to sell 100 ounces of Gold at a delivery price of $K = \$2,000$ per ounce. The contract matures in 3 months.

(a) **Scenario 1:** In 3 months, the spot price of Gold is $S_T = \$1,900$.

- Calculate your total payoff.
- Did you make money or lose money? Explain intuitively why.

Solution:

- **Payoff per ounce:** $K - S_T = 2000 - 1900 = +\100 .
- **Total Payoff:** $100 \text{ oz} \times \$100 = +\$10,000$.
- **Intuition:** You made money because you locked in a selling price of \$2,000, but the market price dropped to \$1,900. You are selling above the market rate.

(b) **Scenario 2:** In 3 months, the spot price of Gold is $S_T = \$2,150$.

- Calculate your total payoff.

Solution:

- **Payoff per ounce:** $K - S_T = 2000 - 2150 = -\150 .
- **Total Payoff:** $100 \text{ oz} \times (-\$150) = -\$15,000$.
- You lost money because you must sell at \$2,000 when the market price is \$2,150.